



Protocol Audit Report

Version 1.0

NuelOnye

May 28, 2026

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May 27, 2026

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Protocol Summary

Protocol does X, Y, Z

Disclaimer

The NuelOnye’s team makes all effort to find as many vulnerabilities in the code in the given time period, but holds no responsibilities for the findings provided in this document. A security audit by the team is not an endorsement of the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the Solidity implementation of the contracts.

Risk Classification

		Impact		
		High	Medium	Low
Likelihood	High	H	H/M	M
	Medium	H/M	M	M/L
	Low	M	M/L	L

We use the CodeHawks severity matrix to determine severity. See the documentation for more details.

Audit Details

The findings decribed in this document correspond with the following commit hash:

```
1 2e8f81e263b3a9d18fab4fb5c46805ffc10a9990
```

Scope

```
1 ./src/  
2 #-- PasswordStore.sol
```

Roles

- Owner: The user who can set the password and read the password.
- Outsiders: No one else should be able to set or read the password.

Executive Summary

Issues found

Severity	Number of issues found
High	2
Medium	0
Low	0
Info	1
Gas Optimizations	0
Total	0

Findings

High

[H-1] Storing the password on-chain makes is visible to anyone, and no longer private

Description: All data stored on-chain is visible to anyone and can be read directly from the blockchain. The `PasswordStore : : s_password` variable is intended to be a private variable and only accessed through the `PasswordStore : : getPassword` function, which is intended to be only called by the owner of the contract

We show one such method of reading any data off chain below

Impact: Anyone can read the private password, severely breaking the functionality of the protocol.

Proof of Concept: (Proof of code)

The below test case shows how anyone can read the password directly from the blockchain.

- ## 1. Create a locally running chain

```
1 make anvil
```

- ## 2. Deploy the contract to the chain

```
1 make deploy
```

- ### 3. Set Password by calling the setPassword function

```
1 cast send <ADDRESS_HERE> "setPassword(string)" "myPassword" --rpc-url
  https://127.0.0.1:8545 --private-key 0
  xac0974bec39a17e36ba4a6b4d238ff944bacb478cbcd5efcae784d7bf4f2ff80
```

- #### 4. Run the storage tool

We use 1 because that's the storage slot of `s_password` in the contract.

```
1 cast storage <ADDRESS_HERE> 1 --rpc-url https://127.0.0.1:8545
```

You'll get an output that looks like this:

[illegible]

You can then parse that hex to a string with:

[illegible]

And you'll get an output of:

```
1 myPassword
```

Recommended Mitigation: Due to this, the overall architecture of the contract should be rethought. One could encrypt the password off-chain, and then store the encrypted password on-chain. This would require the user to remember another password off-chain to decrypt the password. However, you'd also likely want to remove the view function as you wouldn't want the user to accidentally send a transaction with the password that decrypt your password.

[H-2] PasswordStore:setPassword has no access controls, meaning a non-owner could change the password

Description: The `PasswordStore::setPassword` function is set to be an `external` function, however, the natspec of the function and overall purpose of the smart contract is that `This function allows only the owner to set a new password.`

```
1 function setPassword(string memory newPassword) external {
2 @> // @audit - There are no access controls
3     s_password = newPassword;
4     emit SetNetPassword();
5 }
```

Impact: Anyone can set/change the password of the contract. severely breaking the contract intended functionality.

Proof of Concept: Add the following to the `PasswordStore.t.sol` test file

Code

```
1 function test_anyone_can_set_password(address randomAddress) public
2 {
3     vm.assume(randomAddress != owner);
4     string memory expectedPassword = "malicious-password";
5     vm.prank(randomAddress);
6     passwordStore.setPassword(expectedPassword);
7
8     vm.startPrank(owner);
9     string memory actualPassword = passwordStore.getPassword();
10    assertEq(actualPassword, expectedPassword);
11 }
```

Recommended Mitigation: Add an access control conditional to the `setPassword` function.

```
1 if(msg.sender != s_owner){
2     revert PasswordStore_NotOwner();
3 }
```

Informational

[I-1] The PasswordStore::getPassword natspec indicates a parameter that doesn't exist, causing the natspec to be incorrect

Description:

```
1 /*
2  * @notice This allows only the owner to retrieve the password.
3 @> * @param newPassword The new password to set.
4  */
5 function getPassword() external view returns (String memory){}
```

The `PasswordStore::getPassword` function signature is `getPassword()` which the natspec say it should be `getPassword(string)`.

Impact: The natspec is incorrect.

Recommended Mitigation: Remove the incorrect natspec line.

```
1 - * @param newPassword The new password to set.
```